

ASHOM 1

Quantum Equity Model

A Hybrid Quantum-Econometric Stock-Classification Framework

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Version 1.1 - August 2026

Abstract. This paper specifies Ashom 1, a hybrid quantum-classical framework for forecasting short- and long-horizon net benchmark-relative stock returns. Classical systems collect and timestamp market, financial, macroeconomic, geopolitical and ownership information. Official issuer disclosures are the primary source for company fundamentals, named economic relationships and disclosed risks. News may update the event state of a disclosure-supported relationship but may not create the relationship itself. A ten-qubit feature map measures nonlinear similarity, and benchmark-weighted kernel-ridge regression converts that similarity into forecasts. Classification depends on lower forecast bounds plus liquidity, coverage and tail-risk gates. No quantum advantage is claimed unless sealed walk-forward tests beat preregistered classical models after costs.

1. Objective

For stock i , forecast origin t , and horizon h in $\{20, 252\}$ trading sessions, the model predicts future net benchmark-relative return. The 20-session model is primary. The 252-session model activates only when enough non-overlapping annual origins and distinct regimes exist for defensible inference.

$$\text{NXR}_{i,t}^{(h)} = R_{i,t \square t+h} - R_{b,t \square t+h} - \text{TC}_{i,t}^{(h)}$$

The benchmark must be point-in-time and total-return consistent. Estimated transaction costs contain fees, spread and position-size impact using information available at the forecast origin.

2. Disclosure-First Evidence Hierarchy

Ashom 1 separates the source that establishes an economic fact from the source that describes a recent event. This prevents news, model memory or a market-data aggregator from silently replacing a financial statement.

Priority	Source	Permitted role
1	Official issuer disclosures	Identity, statement values, named customers, suppliers, competitors, partners, investments and disclosed risks.
2	Market feeds	Daily traded prices, volume, dividends and splits that filings cannot provide at market frequency.
3	Official macro series	GDP growth, inflation and unemployment with observation year and release lag preserved.
4	Cited news	Recent local, international and geopolitical event tone only. News cannot establish a relationship edge or substitute for a statement.

Saudi equities. The primary company source is the Saudi Exchange issuer announcement or official financial profile. Announced thousands, millions and billions are converted to Saudi riyals. Missing rows remain null.

US equities. The SEC ticker index resolves the issuer and CIK; the latest 10-K, 10-K/A, 20-F or 40-F supplies the official annual filing; and SEC Company Facts supplies standardized XBRL observations. Missing or issuer-specific facts remain null.

The live interface retrieves the latest available filing. Historical estimation must preserve the filing public at each forecast origin. Substituting a later filing into an earlier row creates look-ahead bias even when the document itself is official.

3. Disclosure-Anchored Company Network

Let j be a named company related to target firm i . Ashom 1 creates edge $i \leftrightarrow j$ only when the target firm's official disclosure names j and states an interpretable relationship. The model records the filing URL and a short supporting passage. General model knowledge, an unnamed phrase such as 'a major customer,' and a news-only association are insufficient.

Once an edge is accepted, recent news about company j may update that edge. Each event has tone T_e in $[0,100]$, transmission direction d_e in $\{-1,0,1\}$, filing-supported relationship strength r_e , classifier confidence c_e , source quality q_e , and age a_e in days.

$$w_e = r_e c_e q_e \exp[-\ln(2) a_e / 30]$$

Relationship strength uses $\{0.35, 0.65, 1.00\}$; confidence and source quality use $\{0.50, 0.75, 1.00\}$. The 30-day half-life reduces the influence of older events.

$$N_{i,t} = \text{clip}[50 + 20 \times \sum w_e d_e (T_e - 50) / (50 \sum w_e), 0, 100]$$

- Require at least two distinct events with nonzero direction.
- Remove duplicate headlines and target-company headlines already counted directly.
- Cap the complete network at a 20-point movement from neutral.
- Enter all network events as one aggregate factor to prevent headline-count double counting.

This is a transparent screening index, not an expected return, probability or causal exposure coefficient. In the fitted model, the network shock is centered and standardized inside each training fold, interacted with point-in-time documented exposure, and assigned an estimated coefficient. A filing can verify an edge without quantifying revenue dependence, contract duration or substitution elasticity.

4. Point-in-Time Quantum Feature Vector

$$x_{i,t} = [M, \sigma, DD, \beta, E, F, X, L, I, G]'$$

Qubit	Factor	Point-in-time construction
1	Momentum M	Benchmark-centered 120-20 session momentum.
2	Volatility σ	Annualized volatility from the preceding 20 sessions.
3	Drawdown DD	Maximum drawdown over the preceding 252 sessions.
4	Market beta β	Rolling sensitivity to the frozen market benchmark.
5	Market/network E	Oil and rate sensitivities plus the disclosure-anchored network shock.
6	Fundamentals F	Publication-vintage growth, profitability and quality.

Qubit	Factor	Point-in-time construction
7	Macro exposure X	Macroeconomic conditions interacted with firm exposure.
8	Local perception L	Residualized surprise in local and Arabic-language tone.
9	International I	Residualized surprise in foreign reporting tone.
10	Perception gap G	Lagged foreign ownership multiplied by I-L.

$$G_{i,t} = FO_{i,t-1}(I_{i,t} - L_{i,t})$$

The perception gap preserves the distinction between domestic investors, who may remain invested during a local event, and international investors, who may respond more strongly to foreign coverage. News tone alone is not investor behavior.

$$E_{i,t} = \omega_o E_{i,t}^{oil} + \omega_r E_{i,t}^{rate} + \omega_n N_{i,t}$$

N is the centered, training-fold-standardized network shock. The weights must be frozen or estimated inside the training fold; the display score is never inserted as if it were a known return contribution. Missing required features fail the coverage gate and are not filled with zero or 50.

5. Leakage Control and Quantum Encoding

Within each walk-forward training fold, feature j is standardized, clipped to $[-3,3]$, and converted to an angle. Training means and scales are never estimated with validation or future rows.

$$z_{i,t,j} = \text{clip}[(x_{i,t,j} - \mu_{\text{train},j}) / s_{\text{train},j}, -3, 3]; \theta_{i,t,j} = (\pi/3) z_{i,t,j}$$

Starting from $|0\rangle^{10}$, the feature map applies a Hadamard gate to every qubit. Two data-reuploading repetitions then apply feature-dependent R_Y and R_Z rotations plus ring-connected R_{ZZ} interactions. Entanglement represents nonlinear combinations such as leverage with rate exposure, oil shocks with sector exposure, and foreign ownership with the international-local perception gap.

6. Quantum Fidelity Kernel

$$K_Q(x,z) = |\langle \varphi(x) | \varphi(z) \rangle|^2, 0 \leq K_Q(x,z) \leq 1$$

A value near one indicates similar observations in the quantum feature space. For scalability, each fold selects at most 64 chronological Nyström landmarks without future outcomes. The kernel representation is the vector of similarities between the observation and those landmarks.

7. Hybrid Quantum-Classical Estimator

$$\hat{f}_Q(x) = \hat{\beta}_o + k_M(x)' \hat{\beta}$$

Benchmark-weighted kernel-ridge regression estimates the parameters by minimizing weighted squared net-excess-return errors plus $\lambda \|\beta\|^2$, with review penalty $\lambda = 10^{-3}$. The quantum circuit constructs the similarity kernel; regression, uncertainty, risk controls and governance remain classical.

8. Walk-Forward Estimation

- Retain only observations whose outcomes were fully realized before the evaluation origin.
- Rebuild scaling, perception residualization, the disclosure-anchored company graph, network shocks, exposure interactions and landmarks inside the training fold.
- Fit the quantum model and every classical benchmark on identical rows.
- Produce one untouched out-of-sample forecast.
- Advance to the next non-overlapping forecast origin.

Random train-test splitting is not permitted for the primary result.

9. Forecast Uncertainty

A full-pipeline Rademacher wild-cluster bootstrap changes residual signs at the preregistered firm-origin cluster level, refits the model, and predicts the untouched row. The decision statistic is the fifth percentile of the resulting forecast distribution.

$$L_{i,t}^{(h)} = Q_{0.05} \{ \text{NXR}_{i,t,1}^{(h)}, \dots, \text{NXR}_{i,t,B}^{(h)} \}$$

A percentage of observed inputs is not a confidence interval and cannot substitute for the bootstrap distribution.

10. Classification Rule

Review hurdles are 0.5% for the 20-session lower bound, 3.0% for the 252-session lower bound, and -8.0% for 20-session expected shortfall. Required risk gates retain veto power.

Condition	Classification
Required inputs or uncertainty draws unavailable	Insufficient data
Expected-shortfall or required risk gate fails	Do not invest
Short and long lower bounds pass	Both
Only short lower bound passes	Short-term trading
Only long lower bound passes	Long-term investment
Neither passes, with adequate coverage	Do not invest

11. Illustrative Calculation

11.1 Company-network screen

Suppose a dated annual filing names Firm A as a supplier, giving medium relationship strength $r = 0.65$. Two distinct Firm A events pass the evidence checks.

Event	Tone	Direction	Confidence	Source	Age
Production expansion	80	1	1.00	1.00	5 days
New customer contract	70	1	0.75	0.75	10 days

The event weights are approximately 0.579 and 0.290. Their weighted deviation from neutral is approximately 26.7, giving:

$$N_{i,t} \approx 50 + 20(26.7/50) = 60.7 \approx 61$$

This means moderately favorable network context, not a 61% return or probability. Without the filing-supported Firm A edge, or with only one usable event, the network score remains missing.

11.2 Forecast classification

Quantity	Illustrative value
Expected 20-session net excess return	1.4%
20-session fifth-percentile bound	0.7%
Expected 252-session net excess return	4.2%
252-session fifth-percentile bound	2.0%
20-session expected shortfall	-6.0%

The short lower bound exceeds 0.5%, the long lower bound does not exceed 3.0%, and expected shortfall remains above the -8.0% veto. The result is *Short-term trading*. These values are illustrative and are not a forecast for any named security.

12. Classical Benchmarks and Falsification

The quantum model must be compared on identical sealed rows against benchmark-weighted linear WLS, elastic net, classical RBF-kernel regression, gradient-boosted trees and the transparent deterministic screen. Report forecast error, rank correlation, net long-short return, turnover, expected shortfall, regime calibration, runtime and circuit-evaluation count. Reject the quantum hypothesis if it fails to beat the strongest classical baseline after costs and uncertainty.

Timestamped recommendations from Goldman Sachs, J.P. Morgan and two local Saudi banks remain sealed from feature construction, fitting, hyperparameter selection and hurdle selection. They are opened only for final external comparison after the pipeline is frozen.

13. Current Status

The ten-qubit circuit and quantum fidelity kernel are executable with the local exact-statevector reference implementation. The application now retrieves Saudi Exchange financial disclosures for Saudi issuers and SEC annual filings plus standardized XBRL facts for US issuers. Those documents are the only permitted basis for named company-network edges. Recent news applies only to event transmission along accepted edges. The interface displays the filing authority, document link, supporting passage, related-company events and capped network factor.

These additions improve evidence discipline but do not unlock a validated forecast. Production still requires a frozen point-in-time panel, historical benchmark weights, publication-vintage statements, ownership data, realistic transaction costs, fitted coefficients and the complete uncertainty artifact. Running a quantum circuit is not evidence of quantum advantage.

Source-to-Variable Audit

Model item	Primary authority	Missing-data rule
Financial fundamentals	Issuer filing / official XBRL	Remain null; no news substitution
Company relationship edge	Named filing passage	No passage means no edge
Related-company event	Cited recent news	Needs an accepted filing edge
Price and trading factors	Market feed	Unavailable observations are excluded
Macroeconomic conditions	Official macro series	Preserve year and release lag
Foreign ownership	Timestamped exchange disclosure	Required before perception-gap fitting

Primary Data Authorities

- Saudi Exchange issuer announcements and company profiles - <https://www.saudiexchange.sa/>
- SEC EDGAR filings and submissions data - <https://www.sec.gov/edgar/search/>
- SEC standardized Company Facts API - <https://data.sec.gov/api/xbrl/companyfacts/>
- World Bank World Development Indicators - <https://api.worldbank.org/>

Research Disclaimer

Ashom 1 is a research prototype. Its outputs are not personalized investment advice and do not guarantee future performance.